Huawei Access Node HEB-ACC-40 Hebbal Edge QoS / Congestion Event Troubleshooting SOP

Node: HEB-ACC-40 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_1902

Description

This Standard Operating Procedure (SOP) addresses the troubleshooting of a QoS / Congestion Event, specifically Microbursts, interface tail drops, and QoS queue starvation on the Huawei Access Node HEB-ACC-40 located at the Hebbal Edge site. This failure domain is attributed to the inherent complexities of Quality of Service (QoS) policy enforcement and the interplay between packet forwarding, scheduling, and queuing mechanisms within the node's hardware and software architecture. The Huawei Access Node, being a critical component of the Hebbal Edge site's network infrastructure, plays a pivotal role in ensuring the efficient and reliable transmission of data traffic. The occurrence of QoS / Congestion Events can have far-reaching implications on network performance, packet loss, and overall user experience. Therefore, it is essential to identify and rectify the root causes of such events in a timely and effective manner to prevent network instability and ensure the continued delivery of high-quality services.

Pre-requisite Checks

- Check: Verify that the Hebbal Edge site is accessible and that the Huawei Access Node HEB-ACC-40 is powered on and fully operational., Description: Ensure that all necessary physical and environmental checks have been performed to confirm the node's operational status.

- Check: Confirm the current network topology and configuration of the Hebbal Edge site to identify potential bottlenecks and QoS policy conflicts., Description: Review the current network configuration to identify any potential causes of QoS / Congestion Events.

Potential Root Causes

Cause	Probability
QoS policy conflicts or misconfigurations	High
Hardware or software issues with the Huawei Access Node HEB-ACC-40	Medium
Network congestion or oversubscription at the Hebbal Edge site	Low

Diagnostic Commands & Expected Outputs

```
Command: display queue queue-index 1 queue-type 1, Expected Output: Queue index: 1, Queue type: 1, Current packets: 100, Threshold: 50, Description: This command displays the current queue status, including packet count and threshold values, to diagnose QoS queue starvation.

Command: display interface interface-name Eth1/0/0, Expected Output: Interface Eth1/0/0, administrative status: up, operational status: down, Description: This command checks the operational status of the
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interface to diagnose potential tail drop events.

Command: display qos policy policy-name Policy_1, Expected Output:
Policy name: Policy_1, Policy type: QoS, Priority: 10, Description:
This command displays the QoS policy configuration to diagnose
potential policy conflicts or misconfigurations.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration and update if necessary, Command: undo qos policy policy-name Policy_1, Description: If the QoS policy is found to be misconfigured, update the policy configuration to resolve the issue.

Step 2: Step: Reset the Huawei Access Node HEB-ACC-40, Command: reset system, Description: If the hardware or software issue is found to be the root cause, reset the Huawei Access Node HEB-ACC-40 to its default configuration.

Step 3: Step: Configure QoS queue threshold values, Command: queue queue-index 1 queue-type 1 threshold 100, Description: Configure the QoS queue threshold values to prevent queue starvation.

Post-Check Verification

```
Command: display queue queue-index 1 queue-type 1, Expected Output:
Queue index: 1, Queue type: 1, Current packets: 0, Threshold: 100,
Description: Verify that the QoS queue threshold values have been
successfully configured.

Command: display interface interface-name Eth1/0/0, Expected Output:
Interface Eth1/0/0, administrative status: up, operational status: up,
Description: Verify that the interface is operational and that tail
drop events have been resolved.
```

Roll-back Procedures

- Step: Restore the original QoS policy configuration, Command: undo qos policy policy-name Policy_1, Description: If the fix fails, restore the original QoS policy configuration to prevent network instability.

- Step: Restore the default configuration of the Huawei Access Node HEB-ACC-40, Command: reset system, Description: If the fix fails, restore the default configuration of the Huawei Access Node HEB-ACC-40 to its default state.